NETWORK LAB REPORT

BACKGROUND	
Date:	2026-02-13
Student:	BruceD
Lab Title:	1.3.6 Packet Tracer – Configure SSH
Course:	CCNA SRWE
Device:	Cisco Catalyst Switch (S1)
Management Interface:	VLAN 1 – 10.10.10.2 /24
Workstation Address:	10.10.10.10 /24
Management Network:	10.10.10.0/24
Objective:	– Harden the switch management plane. – Eliminate insecure Telnet access. – Enforce encrypted SSH remote administration. – Implement identity-based authentication. – Persist configuration securely. – Validate behavioral change through controlled testing.

INITIAL STATE ANALYSIS

Baseline Remote Access Behavior

- Telnet session established over TCP port 23 using `telnet 10.10.10.2`.
- Switch prompted for line password authentication rather than username-based login.
- Privileged EXEC access obtained using plaintext `enable password`.
- `show running-config` revealed `no service password-encryption`.
- Credentials visible in readable format within configuration output.
- No hostname-domain pairing configured, preventing SSH initialization.
- No RSA key pair present; SSH process not running.
- VTY lines configured with default transport allowing Telnet.
- `show ip ssh` indicated SSH disabled and prompted RSA key creation, confirming no SSH service readiness.
- Operational note: pipe filtering (e.g., `| include`, `| section`) requires Privileged EXEC (`S1#`) rather than User EXEC (`S1>`).

```
C:\>telnet 10.10.10.2
Trying 10.10.10.2 ...Open


User Access Verification

Password:
S1>enable
Password:
S1#show running-config | section line vty
line vty 0 4
 password cisco
 login
line vty 5 15
 password cisco
 login
S1#show ip ssh
SSH Disabled - version 1.99
%Please create RSA keys (of atleast 768 bits size) to enable SSH v2.
Authentication timeout: 120 secs; Authentication retries: 3
S1#show running-config | include ip domain-name
S1#show crypto key mypubkey rsa

S1#show running-config | include service password-encryption
no service password-encryption
S1#show running-config | include enable password|password
no service password-encryption
enable password cisco
 password cisco
 password cisco
S1#
```

```
S1#show running-config
Building configuration...

Current configuration : 1150 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1
!
!
enable password cisco
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
```

Security Risk Assessment

- Telnet transmits credentials in cleartext over TCP port 23.
- An attacker within the same broadcast domain could capture packets using Wireshark and recover credentials.
- Lack of cryptographic identity prevents verification of device authenticity.
- Plaintext enable password storage increases risk if configuration backup is exposed.
- Absence of encryption compromises confidentiality and session integrity.
- Management plane susceptible to credential interception and replay attacks.
- Shared line password model reduces accountability (no per-user identity), complicating auditing and incident response.

CONFIGURATION PROCEDURE

Hardening Steps Executed

- Entered global configuration mode using `conf t`.
- Executed `service password-encryption` to obfuscate existing passwords.
- Verified encryption using `show running-config | include password`.
- Configured domain name with `ip domain-name netacad.pka`.
- Generated RSA keys using `crypto key generate rsa` (1024-bit modulus selected).
- Observed system message confirming SSH activation.
- Created local account using `username administrator secret cisco`.
- Entered VTY configuration context via `line vty 0 15`.
- Applied `login local` to enforce username authentication.
- Applied `transport input ssh` to disable Telnet transport.
- Removed legacy line password using `no password`.
- Exited configuration mode and saved changes using `copy running-config startup-config`.
- Post-change validation performed using targeted `show` commands to confirm each control took effect.

```
S1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#service password-encryption
S1(config)#end
S1#show running-config | include password
service password-encryption
enable password 7 0822455D0A16
 password 7 0822455D0A16
 password 7 0822455D0A16
S1#
```

Command Purpose and Security Rationale

- `service password-encryption` converts plaintext line/enable passwords into Type 7 obfuscated values in configs.
- Note: Type 7 is reversible obfuscation (deterrent only), and does not provide strong cryptographic protection for stored credentials.
- `ip domain-name` is required to generate RSA keys; without it, SSH cannot initialize.
- `crypto key generate rsa` creates the asymmetric key pair used for SSH encryption and device identity.
- Modulus selection impacts strength and performance; 1024-bit was used for this lab (production commonly uses 2048-bit+ where supported).
- `username ... secret` stores credentials as a one-way hash (Type 5 in Packet Tracer) instead of plaintext.
- `login local` forces VTY authentication against the local username database (not shared line passwords).
- `transport input ssh` restricts inbound VTY management to SSH only, disabling Telnet as an access method.
- `no password` removes the legacy line password fallback, preventing authentication from reverting to shared password mode.
- `copy running-config startup-config` commits hardening changes to NVRAM so settings persist after reload.

Cryptographic Initialization

- RSA key pair generated to provide asymmetric encryption capability.
- Public key used during SSH handshake for identity verification.
- Hostname-domain combination embedded into key generation process.
- SSH service initialized automatically after key generation.
- SSH version 1.99 (Packet Tracer simulated) supports SSHv2 protocol features.
- Administrator credentials stored as Type 5 MD5 hash in configuration.
- Device identity and secure channel establishment now depend on the generated RSA key material.

OPERATIONAL VALIDATION

Remote Access Behavior After Hardening

- Terminated Telnet session and exited to workstation command prompt.
- Attempted Telnet reconnect; connection immediately closed by switch.
- Verified Telnet no longer accepted on TCP port 23.
- Initiated SSH session using `ssh -l administrator 10.10.10.2`.
- Prompted for password linked to local username database.
- Successfully authenticated and accessed User EXEC mode.
- Elevated to Privileged EXEC mode using `enable`.
- Confirmed encrypted channel by observing SSH protocol behavior and successful authentication.
- Authentication model validated as per-user identity rather than shared line password.

Configuration Verification and Persistence

- Executed `show running-config` to confirm SSH and username entries present.
- Executed `show startup-config` to verify configuration written to NVRAM.
- Confirmed `service password-encryption` persisted after save.
- Observed RSA-related entries increasing configuration size.
- Verified VTY section contained `login local` and `transport input ssh`.
- Confirmed absence of Telnet in VTY transport configuration.
- Verified SSH status displayed as enabled following key generation and VTY hardening.

```
C:\>telnet 10.10.10.2
Trying 10.10.10.2 ...Open

[Connection to 10.10.10.2 closed by foreign host]
C:\>ssh -l administrator 10.10.10.2

Password:



S1>
```

```
S1#show running-config | include service password-encryption
service password-encryption
S1#show crypto key mypubkey rsa
% Key pair was generated at: 23:33:37 UTC February 28 1993
Key name: S1.netacad.pka
 Storage Device: not specified
 Usage: General Purpose Key
 Key is not exportable.
 Key Data:
 0000639a  0000528d  000033cc  00005df7  00002d31  00007003  000008fc  000000f3
 00005add  0000285e  00000c0f  00002291  000067bc  000055d4  00000bb0  00007ee1
 00006770  00002613  00000528  00004b1a  0000731f  00006332  00001980  6631
% Key pair was generated at: 23:33:37 UTC February 28 1993
Key name: S1.netacad.pka.server
Temporary key
 Usage: Encryption Key
 Key is not exportable.
 Key Data:
 000067be  000048e8  00005de6  000077aa  00005990  000039fc  00000b53  000070af
 000060e9  000069fc  0000516a  0000235f  00002438  00004f32  000050a2  00006468
 00004344  000070d4  00007fa7  00006ad8  000007b6  0000657b  00007ffe  6230
S1#show running-config | include username
username administrator secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
S1#show running-config | section line vty
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
S1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
S1#
```

Management Plane Behavior Change

- Remote authentication shifted from shared password model to identity-based model.
- Session confidentiality now enforced via encrypted transport (TCP 22).
- Integrity ensured through SSH message authentication mechanisms.
- Administrative access now dependent on cryptographic identity validation.
- Attack surface for credential harvesting significantly reduced.
- Administrative actions can be tied to a specific username, improving accountability.

ANALYSIS SUMMARY

Security Improvements Achieved

- Management access migrated from plaintext Telnet to encrypted SSH protocol.
- Credential exposure risk substantially reduced.
- Device cryptographic identity established via RSA key pair.
- Local username database replaced shared line authentication model.
- Passwords no longer stored in readable plaintext.
- Configuration persistence ensures security controls survive reload events.
- Management access control enforces a secure transport baseline suitable for enterprise hardening.

Operational Outcome

- Switch management plane transitioned from insecure baseline to encrypted administrative model.
- Observed system behavior confirms successful protocol enforcement change.
- Remote administrative access now aligns with secure enterprise networking standards.
- Configuration state reflects consistent and validated hardening implementation.
- Control effectiveness demonstrated through configuration-state checks and behavioral testing.

Control / Check	Before	After
Telnet accepted	✓	×
SSH enabled	×	✓
VTY transport	telnet/ssh	ssh-only
Authentication model	line password	local user
Password obfuscation	off	on
Persistence (startup-config)	not saved	saved

Verification Evidence Collected

- Domain binding (SSH prerequisite): `show running-config | include ip domain-name`.
- Password obfuscation active: `show running-config | include service password-encryption`.
- RSA key material present: `show crypto key mypubkey rsa`.
- Local user database entry exists: `show running-config | include username`.
- VTY restricted to encrypted access: `show running-config | section line vty`.
- SSH status and parameters: `show ip ssh`.
- Behavioral proof: Telnet rejected; SSH authentication succeeded using local credentials.
- Persistence proof: `show startup-config` reflected saved hardening controls.
- SSH dependency note: prior to RSA key generation, `show ip ssh` remained disabled and required RSA keys.
- CLI note: Packet Tracer filtering may be inconsistent; use full `show startup-config` if an include is rejected.

Control-to-Risk Mapping

- `transport input ssh` → removes plaintext credential exposure via Telnet (TCP/23).
- RSA key generation → establishes cryptographic device identity and secure session negotiation.
- `login local` + local user `secret` → enforces identity-based authentication per user.
- `service password-encryption` → reduces risk of readable password disclosure in configs/backups.
- `copy running-config startup-config` → prevents rollback to insecure state after reload.
- `no password` on VTY → removes shared line-password fallback, eliminating ambiguous authentication.

Residual Risk and Recommended Next Steps

- Recommend migrating legacy `enable password` to `enable secret` for stronger protection.
- Recommend enforcing SSHv2 explicitly with `ip ssh version 2`.
- Recommend 2048-bit RSA keys for production environments (used 1024-bit modulus).
- Recommend VTY hardening (timeouts / rate limiting) to reduce brute-force and idle-session risk.
- Recommend applying a management ACL to restrict SSH to trusted admin hosts/subnet only.
- Environment note: Packet Tracer reports SSH as version 1.99; validate SSHv2 policy/crypto standards in production.
- Scope boundary: This lab validated SSH enablement and VTY transport hardening; ACLs/timeouts and broader policy are follow-on controls.